Problem Pack: Pythagorean Theorem, MATHEMATICS 8 - UNIT 2 REVIEW

Question 12

Problem: 12. For the right triangle shown: (a) identify the hypotenuse, and (b) write the Pythagorean relationship using the side labels.

Student Work:

a) c
b) a^2 + b^2 = c^2

Feedback

Question 13

Problem: 13. A right triangle has legs of 5 cm and 5 cm. Find the hypotenuse. Leave a non-whole-number answer in simplest radical form.

Student Work:

c^2 = 5^2 + 5^2 = 25 + 25 = 50 c = √50 = √25 x 2 = 5√2

Feedback

Question 14

Problem: 14. A right triangle has a hypotenuse of 7 cm and one leg of 4 cm. Find the other leg. Leave a non-whole-number answer in simplest radical form.

Student Work:

x^2 + 4^2 = 7^2 x^2 = 49 - 16 = 33 x = √33

Feedback

Question 15

Problem: 15. To find the unknown leg a of a right triangle with hypotenuse 25 and one leg 20, a student wrote a = √25^2 + 20^2. Explain the error and find the correct value of a.

Student Work:

a^2 + 20^2 = 25^2 a^2 = 625 - 400 = 225 a = 15

Feedback

Question 16

Problem: 16. Determine whether each set of side lengths can form a right triangle. Justify each answer using the Pythagorean theorem. a) 7, 24, 25 b) 5, 6, 8 c) 28, 45, 53

Student Work:

a) 7^2 + 24^2 = 49 + 576 = 625 = 25^2

b) 5^2 + 6^2 = 25 + 36 = 61
c) 28^2 + 45^2 784 + 2025 = 2809

Feedback

In question 16, you forgot to check part c by seeing if the sum matches the square of the longest side. You stopped after adding the two smaller squares together, but you need to check if that total equals the square of 53. Try finishing the comparison to see if it makes a true statement.